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Angular scattering of protons through ultrathin graphene foils: Application for time-of-flight instrumentation

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Angular scattering of protons through ultrathin graphene foils: Application for time-of-flight instrumentation

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ABSTRACT

Space plasma instruments often rely on ultrathin carbon foils for incident ion detection, time-of-flight (TOF) mass spectrometry, and ionization of energetic neutral atoms. Angular scattering and energy loss of ions or neutral atoms in the foil can degrade instrument performance, including sensitivity and mass resolution; thus, there is an ongoing effort to manufacture thinner foils. Using new 3-layer graphene foils manufactured at the Los Alamos National Laboratory, we demonstrate that these are the thinnest foils reported to date and discuss future testing required for application in space instrumentation. We characterize the angular scattering distribution for 3–30 keV protons through the foils, which is used as a proxy for the foil thickness. We show that these foils are ~2.5–4.5 times thinner than the state-of-the-art carbon foils and ~1.6 times thinner than other graphene foils described in the literature. We find that the inverse relationship between angular scattering and energy no longer holds, reaffirming that this may indicate a new domain of beam–foil interactions for ultrathin (few-layer) graphene foils.

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I. INTRODUCTION

Ultrathin carbon foils are commonly used in space plasma instrumentation, for example, time-of-flight (TOF) ion mass spectrometers (e.g., NASA Van Allen Probes/HOPE, Cluster/CODIF, JUNO/JADE) (Funsten *et al.*, 2013; Möbius *et al.*, 1998; Rème *et al.*, 1997; and McComas *et al.*, 2017) and energetic neutral atom (ENA) imagers (e.g., NASA IBEX/IBEX-Hi, TWINS) (Funsten *et al.*, 2009; McComas *et al.*, 2012). These instruments exploit several useful properties of the interaction physics of atoms and molecules as they traverse an ultrathin foil. First, the interaction can generate secondary electrons from both the front and exit surfaces of a foil. Detection of these secondary electrons allows indirect measurement of the incident particle—for example, TOF spectrometry can provide the time at which the particle traversed the foil (Gloeckler and Hsieh, 1979). Second, the foil can modify the charge state distribution of atoms, which is an essential process for ionization of energetic neutral atoms so that they can be subsequently electrostatically analyzed

by standard ion optics (Funsten *et al.*, 2009; McComas *et al.*, 1991). Third, foils dissociate molecules into atomic constituents that can be subsequently analyzed (McComas *et al.*, 1990; Nordholt *et al.*, 2013; Young *et al.*, 2007; and Funsten *et al.*, 1994).

However, the interaction of atoms and molecules with ultrathin foils can introduce effects, particularly angular scattering and energy loss, which degrade the accuracy and precision of the measurement of the particle energy or speed. Employing a thinner foil reduces these undesirable effects (McComas *et al.*, 1990), and there is no evidence that the useful properties of foils (secondary electron emission, charge-state modification, and molecular dissociation) are degraded by thinner foils. For example, the generation of secondary electrons from a transiting incident ion or neutral atom has been shown to be independent of the foil thickness (Allegrini *et al.*, 2003).

Ultrathin carbon foils have been successfully implemented in many foil-based instruments (Wüest, 1998; McComas *et al.*, 2004; and Allegrini *et al.*, 2016). In a conventional application, the

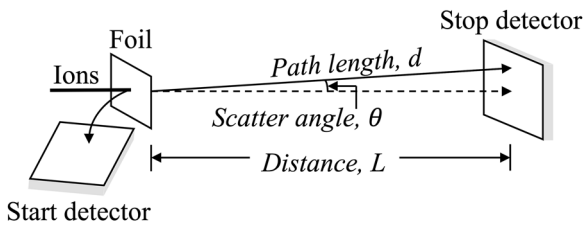


FIG. 1. Schematic of a linear, foil-based TOF measurement used in space plasma mass spectrometry. Ions enter from the left and are incident on a carbon foil (note: this diagram assumes that ions enter perpendicular to the foil). The secondary electrons generated at the foil are measured by the start detector. As the incident ions traverse the foil, the ions scatter with some angle θ and are then detected by the stop detector located a distance L from the foil.

time-of-flight (TOF) measurement follows an electrostatic analyzer, which allows a narrow bandpass of energies into the TOF. Figure 1 shows the basic approach for TOF measurement in a space plasma mass spectrometer. An ion traverses the foil and generates secondary electrons that are electrostatically steered into a “start” detector to start a timer. The ion travels through the field-free drift region and is detected by the “stop” detector, which stops the timer. Ions undergo angular scattering of angle θ when traversing the foil, which results in a traveled path length of $d = L/\cos(\theta)$. The measured time-of-flight t over the ion path length in the drift region yields the ion speed, t/d . With knowledge of the ion energy E that enters the TOF subsystem, the ion mass m is

$$m = 2E \left(\frac{t}{d} \right)^2. \quad (1)$$

Propagating the error through Eq. (1) results in the first order approximation (Taylor, 1982) of the mass resolution,

$$\frac{\delta m}{m} \approx \frac{1}{m} \left(\left| \frac{\partial m}{\partial E} \right| \delta E + \left| \frac{\partial m}{\partial t} \right| \delta t + \left| \frac{\partial m}{\partial d} \right| \delta d \right) = \frac{\delta E}{E} + 2 \left(\frac{\delta t}{t} + \frac{\delta d}{d} \right). \quad (2)$$

However, in a flight instrument, the path length d traveled by each ion is usually not measured. Instead, laboratory measurements of the scattering angle θ for an ensemble of particles are required to characterize the instrument response. The distribution of scattering angles of incident ions introduces uncertainty into the path length d , and energy loss of the ion in the foil introduces uncertainty in knowledge of the ion energy E . Reducing the ion scattering angle by using thinner foils, thus, improves the mass resolution of the instrument.

Conventionally, ultrathin carbon foils used in space instrumentation are made from arc-evaporation of graphite and mounted on a support grid (e.g., 333 lines/in. Ni grid). The thinnest foils used in space have an areal density of $\sim 1.66 \mu\text{g cm}^{-2}$ (Funsten et al., 1994), which corresponds to a thickness of $\sim 80 \text{ \AA}$ or more assuming a density $\leq 2.0 \text{ g cm}^{-3}$ (Bhattara et al., 2017). Thinner carbon foils typically have a substantial fraction of holes and cannot be used in flight instruments.

Recently, ultrathin graphene foils have shown potential for use in space instrumentation (Ebert et al., 2014; Allegrini et al., 2014). The base structure of graphene is a single-atom thick carbon

sheet with a rigid 1D hexagonal honeycomb lattice structure (Geim and Novoselov, 2007); this makes graphene the thinnest known material (Geim, 2009) and is the basis for its superior mechanical strength (Lee et al., 2008) compared to amorphous carbon foils. This allows a few layers of graphene to span the open areas of a grid (Yamaguchi et al., 2018). Other properties important for use as foils in space plasma mass spectrometers include ultra-high electrical conductivity (Novoselov et al., 2004; Dean et al., 2010) to dissipate net charge deposited in the foil and high thermal conductivity (Balandin et al., 2008) to dissipate thermalized kinetic energy transferred from the particle to the foil.

In this study, we investigate the angular scattering of protons through ultrathin (3-layer, thickness of $\sim 20 \text{ \AA}$) graphene foils. We use protons for two important reasons: (1) protons are the standard ion species for angular scattering measurements (e.g., Högberg and Norden, 1970; Funsten et al., 1993; 1994; and Ebert et al., 2014) and (2) protons are one of the most abundant species encountered in space plasma environments (Lennartsson et al., 1979; Lennartsson and Shelley, 1986; Kistler and Mouikis, 2016; and Fernandes et al., 2017). We measure the scattering half-angles, conventionally used as a proxy for foil thickness (Funsten et al., 1992), of 3–30 keV protons. The angular scattering measurements through our graphene foils are compared to previously characterized graphene foils (Ebert et al., 2014), which have not been flown before, and flight-proven conventional carbon foils (Funsten et al., 1993). We show that the Los Alamos developed graphene foils are the thinnest reported to date. Additional testing is needed to implement these graphene foils in space instruments, as detailed in Sec. V. We then compare the angular scattering measurements to Meyer’s theory, which is commonly used to describe foil–beam interactions, and show that graphene foils are in a new domain of physics.

II. EXPERIMENTAL SAMPLE, APPARATUS, AND DATA

A. Synthesis of graphene foils

Graphene was synthesized at the Los Alamos National Laboratory via chemical vapor deposition (CVD) using methane gas on a copper substrate at 1000°C . The low solubility of carbon in copper self-limits the synthesized graphene to monolayer thickness (Li et al., 2009). Poly(methyl methacrylate) (PMMA) was then spin-coated over the graphene monolayer for use as a mechanical support for transfer. The copper substrate was subsequently chemically etched, and the graphene and PMMA sheet was transferred onto a SiO_2/Si substrate for stacking of monolayers. The transfer process was repeated 3 times to achieve 3 monolayers ($\sim 5 \text{ \AA}$ per monolayer), which has been found to have sufficient mechanical stability to span a 3 mm diameter hole (Yamaguchi et al., 2018). After the transfer of each monolayer, the PMMA support layer was removed using an acetone bath followed by thermal annealing in vacuum, which is critical for complete removal of residual PMMA that is known to be present even after acetone bath cleaning (Ishigami et al., 2007). Atomically cleaned 3-layer graphene foils, which are 3 mm in diameter, were finally transferred to 2000 lines/in. (lpi) Ni mesh grids (an open area ratio of 36% and a wire thickness of $5 \mu\text{m}$) for the ion scattering measurements. This grid size was selected to enhance the coverage yield of graphene foils. Two graphene foils were characterized in this study (foil No. 1 and foil No. 2).

B. Apparatus used to measure scatter distribution

Ion scattering measurements were conducted at the Los Alamos Space Plasma Instrumentation Calibration Facility (Funsten *et al.*, 2013). The ion accelerator is capable of generating a monoenergetic (<2 eV energy spread) singly charged ion beam spanning 1–60 keV/e, where e is the fundamental electron charge. The monoenergetic ion beam was magnetically mass-selected to produce a proton beam. During this study, the pressure in the vacuum chamber was $\sim 1 \times 10^{-7}$ Torr, which is typical for testing and calibrating space plasma instruments using such foils.

Figure 2 schematically shows the experimental apparatus used to measure the angular scattering of the proton beam through graphene foils. The proton beam was incident on the grounded beam-defining aperture (A_1 : 0.2 mm diameter), which defined the size of the beam that was incident on the foil. The proton beam then passed through a 5.5 mm diameter aperture (A_2). Aperture A_2 was biased to +20 V to collect secondary electrons created by the beam at A_1 or at the graphene foil. After passing through apertures A_1 and A_2 , the proton beam was incident on the grounded graphene foil mount plate. The mount plate included three identical 3 mm diameter holes spaced 12.7 mm apart, a distance sufficient to ensure that the beam only transits one hole at a time and thus there was no cross-contamination of the beam between adjacent holes. Two graphene foils assessed in this study were placed over two of the holes, and the third hole was left empty to enable a baseline measurement of the ion beam spatial profile at the detector with no graphene foil present. The mount plate was attached to a linear translational stage to allow testing of each foil without breaking vacuum. The ion beam that traversed the empty hole or graphene foil was detected and imaged using a Quantar Technology, Inc., 3395-001/SE imaging microchannel plate (MCP) detector. This MCP provides a two-dimensional, 1024×1024 pixel (39×39 mm) image of detected counts, where the distance between adjacent pixels was measured as (0.039 ± 0.01) mm. The foils were located at a distance $L = (37 \pm 2)$ mm from the MCP. The front of the MCP was biased to -100 V, resulting in an electric field of ~ 2.7 V/mm between the foil and front of the MCP to optimize the imaging resolution of the MCP

detector (Funsten *et al.*, 1996). The back of the MCP was biased at +2.0 kV, while the position sensitive anode was biased at +2.1 kV.

For each measurement, the experimental apparatus allowed precise determination of the proton beam location within a graphene foil area and thus ensured that the proton beam interacted with the same part of each graphene foil. Probing the same part of a graphene foil for all measurements minimizes the uncertainty associated with potential thickness variation across the graphene foils. Uniformity of the graphene foils was evaluated by calculating the scattering half-angle at five different positions on graphene foil No. 1 using a 10 keV proton beam; all five measurements agree within $\leq 7\%$ using the 2D Gaussian fitting method described in Sec. III A.

C. Description of acquired data

The scattering distributions were measured for proton beams of energies 3, 5, 7, 10, 20, and 30 keV through two graphene foils each with a diameter of 3 mm. For each scattering distribution measurement, at least 5000 counts were accumulated with typical values of $\sim 12\,000$ counts. Figure 3 shows the scattering distribution of 5, 10, and 30 keV protons (note: these scattering distribution images represent a small portion, $\sim 0.27\%$, of the entire MCP area). The data shown in Fig. 3 are representative of data taken at the other energies. As expected, the scattering distribution is larger at 5 keV than at 30 keV, indicating that lower energy ions scatter more through foils than higher energy ions.

Before analyzing the data, the detector background as well as beam divergence and finite aperture effects must be removed from the measured scattering distribution. An hour-long background was acquired by operating the MCP detector with no proton beam present. The background measurement was uniform and insignificant; when normalized to the time over which a scattering distribution was acquired, the background counts corresponded to $\sim 0.26\%$ of the total counts accumulated at 10 keV within the region of interest. A background measurement was taken before and after conducting this study to confirm that the MCP was not damaged in the process and that no “hot spots” were developed during the measurements. Additional measurements were conducted to characterize

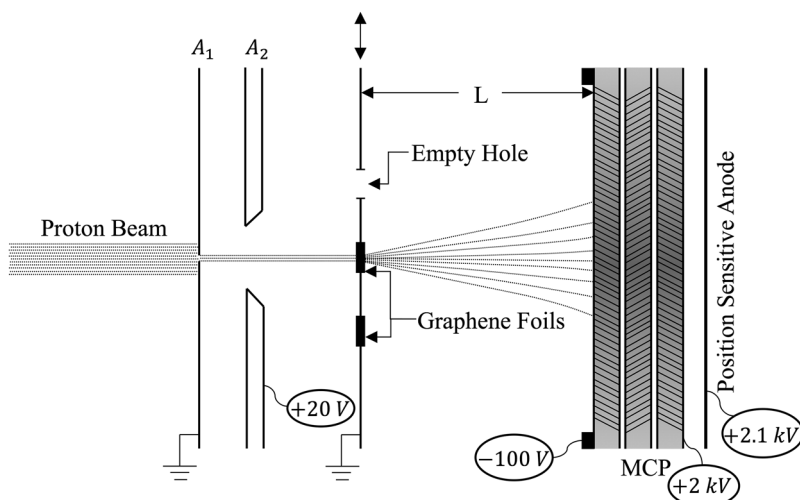


FIG. 2. Diagram of the experimental apparatus. The proton beam enters from the left. Aperture A_1 , 0.2 mm in diameter, defines the beam size, while aperture A_2 , 5.5 mm in diameter, is biased to +20 V to collect secondary electrons generated at aperture A_1 and the front surface of the foil. The graphene foils and empty hole are attached to a translational stage. The beam passes through the graphene foil, and the scattering distribution is measured on the imaging microchannel plate (MCP) detector located at a distance L , 37 mm, from the graphene mount plate.

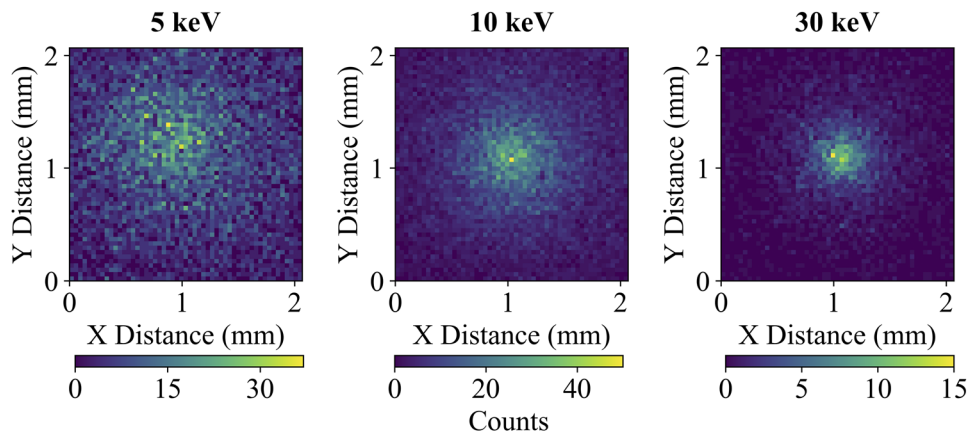


FIG. 3. Scattering distribution of protons that traversed an ultrathin graphene foil. Each column, from left to right, shows the raw scattering distribution measured at the MCP through graphene foil No. 1 for proton energies of 5, 10, and 30 keV. There are 7779, 17 120, and 5523 counts accumulated at 5, 10, and 30 keV, respectively. The single bright pixels are plot outliers.

the ion scattering due to the 2000 lpi grid. Comparison between measurements through the empty hole and measurements through the grid (with no foil) show negligible difference; thus, we conclude that measured ion scattering distributions through the graphene foils are unaffected by the 2000 lpi grid. Section III discusses the data processing techniques used to determine the half-width of the measured scattering distribution.

III. DATA ANALYSIS

The measured scattering distribution is a convolution of three effects: angular divergence of the ion beam that is an intrinsic property of the beam, the size of the finite beam defining aperture, and the angular scattering distribution induced by the passage of the ion beam through the foil. The first two effects can be considered as a source term at the input surface of the foil and can be quantified by measuring the unscattered ion beam that traverses the empty hole in the foil mount plate, i.e., not covered with graphene. The resulting beam observed at the MCP detector represents the projection of the finite aperture area and beam divergence at the imaging plane and thus represents an empirical point spread function (PSF). This empirical distribution, which had a maximum width of 0.24 mm at the MCP detector, was deconvolved from the measured scattering distributions from the graphene foils using a 2D Richardson–Lucy deconvolution algorithm (Richardson, 1972). The 2D deconvolution enables recovery of the “true” scattering distribution for an idealized case of an infinitely small aperture and an ion beam with no angular divergence. Using the deconvolved scatter distribution, the resulting data were then processed to determine the half-width of the scattering distribution.

We characterize the scatter distribution by a single parameter, the scattering half-angle $\psi_{1/2}$,

$$\psi_{1/2} = \tan^{-1}\left(\frac{r_{1/2}}{L}\right), \quad (3)$$

where $r_{1/2}$ is the radial distance from the center of the distribution to the half-width at half maximum (HWHM) of the ion scattering distribution incident on the MCP and L is the distance between the foils and MCP.

Two techniques were used to determine the HWHM ($r_{1/2}$) and the corresponding scattering half-angle ($\psi_{1/2}$): (1) 2D Gaussian fitting and (2) the annular bin method. We first present the analysis using these two techniques, followed by discussion and comparison of the analysis methods. We demonstrate that the 2D Gaussian fitting method is the most robust analysis technique for characterizing the central angular scattering peak and include the annular bin method for direct comparison with previous assessment of graphene foils.

A. 2D Gaussian fitting method

The central peak of the angular scattering distribution, which contains most of the detected counts, is expected to be well-represented by a cylindrically symmetric Gaussian distribution (Hanle and Kleinpopp, 1979). Thus, the deconvolved scattering distribution is naturally fit to a weighted 2D Gaussian distribution described by

$$f(x, y) = Ae^{-\left(\frac{(x-x_0)^2}{2\sigma_x^2} + \frac{(y-y_0)^2}{2\sigma_y^2}\right)} + B, \quad (4)$$

where A is the maximum amplitude, x_0 and y_0 are the horizontal and vertical centers of the distribution on the MCP detector, respectively, σ_x and σ_y are the corresponding standard deviations of the distribution from which the scattering half-angle $\psi_{1/2}$ is derived, and B is the vertical offset or signal baseline. The relative difference between σ_x and σ_y , which should be the same, provides a measure of the error in the scattering half-angle. The Gaussian fits are weighted by the number of counts in each pixel of the deconvolved scattering distribution. Figure 4 shows the resulting deconvolved scattering distribution, contours of constant amplitude from the fitted 2D Gaussian, and the fit residuals for 5, 10, and 30 keV proton beams through foil No. 1. The fit parameters σ_x and σ_y are converted to a HWHM ($r_{1/2}$) in both directions and then used to calculate the scattering half-angles $\psi_{1/2,x}$ and $\psi_{1/2,y}$ using Eq. (3).

B. Annular bin method

The annular bin method is used to directly compare the scattering half-angles to the previously published results through graphene foils (Ebert *et al.*, 2014). For this technique, the deconvolved

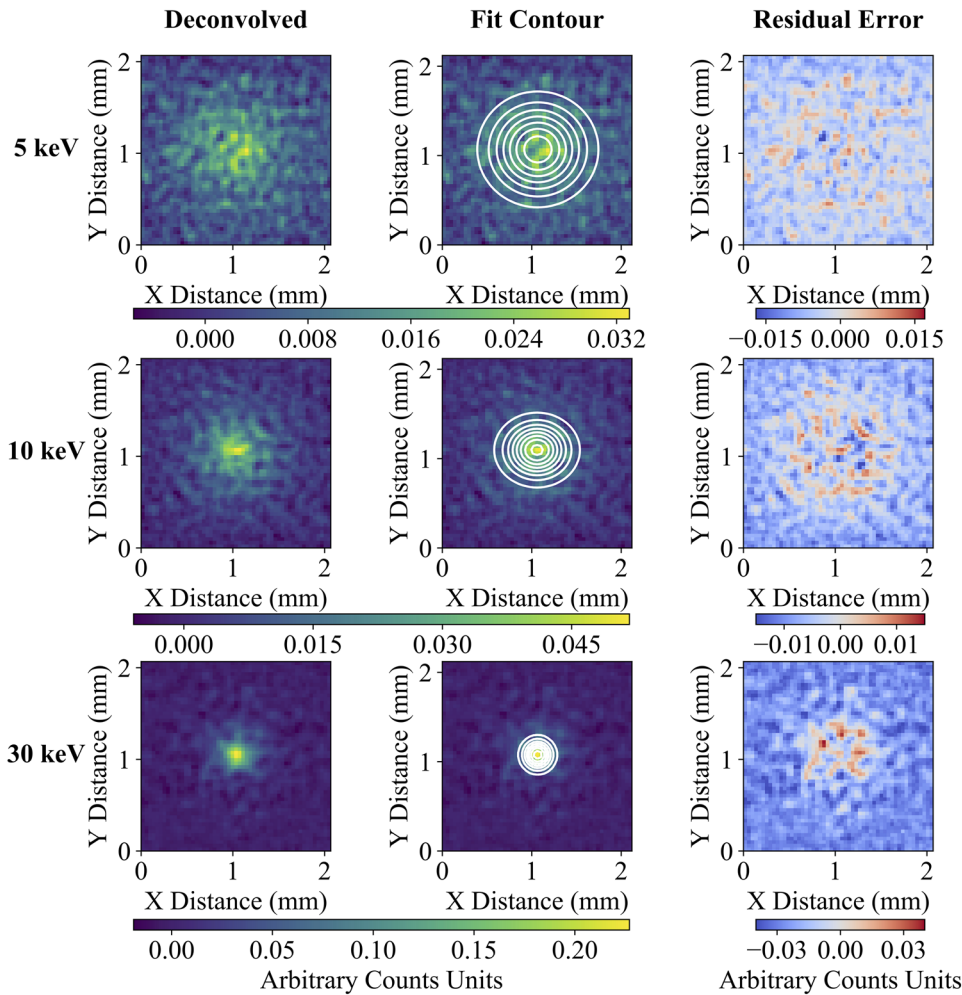


FIG. 4. 2D Gaussian fits to the raw scattering distributions for 5 keV (top row), 10 keV (middle row), and 30 keV protons (bottom row) through graphene foil No. 1. The three columns, from left to right, show the resulting distribution after deconvolving the proton beam, contours of constant amplitude of the weighted 2D Gaussian fit, and the fit residuals.

scattering distribution was first rescaled to have twice as many pixels as the original image to obtain a finer resolution. Then, each pixel in the rescaled deconvolved scattering distribution was mapped to its corresponding scattering angle using Eq. (3), where $r_{1/2}$ is the distance from the center of the scattering distribution to the center of each pixel. The data were then binned into annular rings centered on the maximum of the scattering distribution. By rescaling the image, more accurate circles can be drawn from square pixels; the scattering half-angle is, therefore, less affected by variability in the angular widths. The location of the maximum is a free parameter that is not used in the fit, so its choice can introduce significant errors; for this study, the peak location was determined using the 2D Gaussian fit to the data. The angular limits of each annular bin were defined by $\psi_m \pm \psi_m/2$, where ψ_m is the scattering angle associated with the center radius of each annulus and $\Delta\psi$ is the angular width of each annulus. Within each annulus, the total number of counts was normalized by the annulus solid angle. The angular width $\Delta\psi$ was selected to contain a minimum number of counts such that the statistical counting error is uniform across the annuli. For this study, a uniform angular width of 0.1° was used for all annuli, and the total counts in each

annulus as a function of ψ_m was then empirically fit to a weighted Kappa distribution,

$$f(\psi) = A \left(1 + \frac{\psi_m^2}{\sigma^2 \kappa} \right)^{-\kappa} + B, \quad (5)$$

where A is the maximum amplitude, σ is related to the width of the scatter distribution, κ is related to the magnitude of the distribution's tail, and B is the vertical offset. The counts are used as the weights for the kappa distribution fit. The scattering half-angle for this distribution is $\psi_{1/2} = \sigma \sqrt{\kappa(\sqrt{2} - 1)}$. Figure 5 shows, in columns proceeding from left to right, the rescaled image with the annular bins, the resulting normalized Kappa distribution and fit, and the residual errors.

C. Comparison of analysis techniques

The annular bin technique is sensitive to two free parameters: the angular width and center of the annular rings. When increasing the angular width by 0.05° ($\Delta\psi = 0.15^\circ$) and leaving the center of the annular rings unchanged, the calculated scattering half-angle

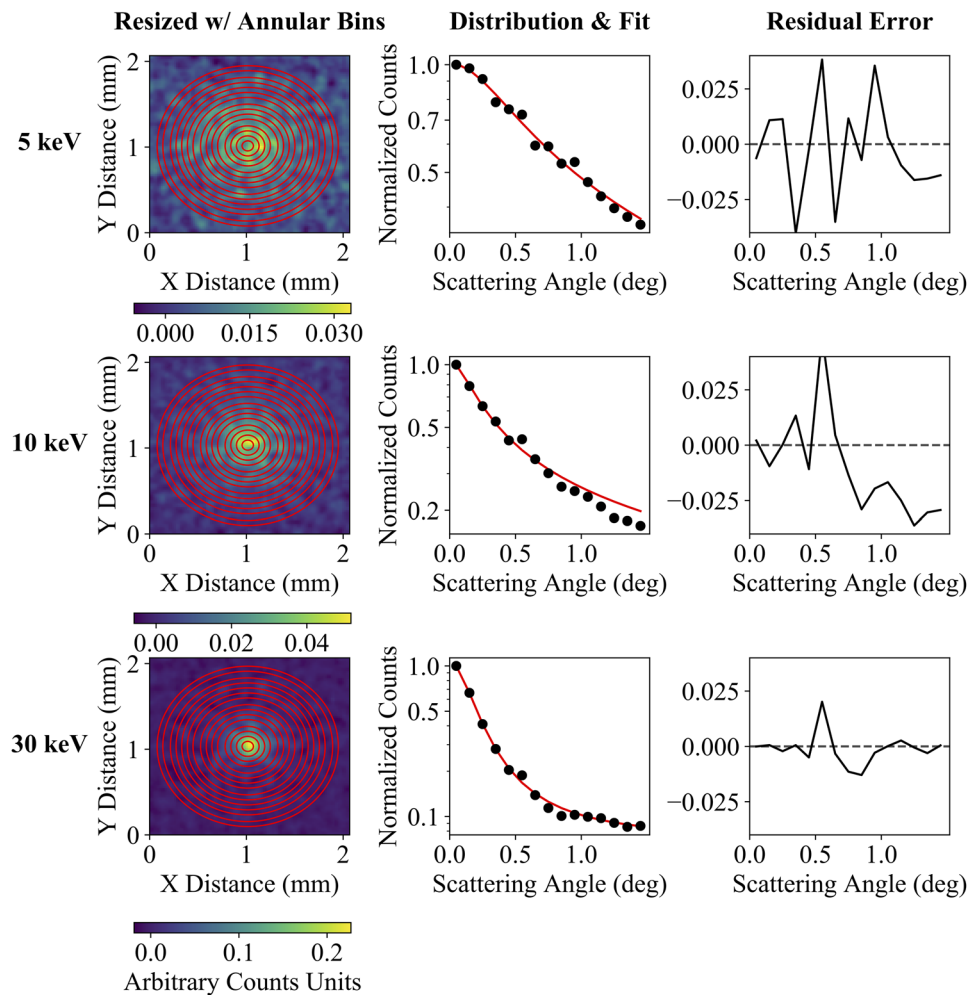


FIG. 5. Annular bin method for 5 keV (top row), 10 keV (middle row), and 30 keV protons (bottom row) through graphene foil No. 1. The three columns, from left to right, show the rescaled image with the 15 annular rings (red circles) with a width of 0.1° and the center based on the result of the Gaussian fit method, the normalized (to unity) count distribution as a function of scattering angle (black dots) with a weighted Kappa fit (red curve), and the residual error from the weighted Kappa fit.

changes by an average of 17%. However, the scattering half-angles are not as sensitive to the center of the annular rings—when moving the center of the annular rings by ~ 0.195 mm (corresponding to $\sim 0.31^\circ$) and leaving the annulus width unchanged ($\Delta\psi = 0.1^\circ$), the scattering half-angle changes by an average of 8%. As shown in Fig. 5, moving the center of the annular rings by 0.195 mm clearly puts the selected peak off the “true” peak of the scattering distribution.

The 2D Gaussian fitting method is inherently designed to center itself based on the measured peak of the distribution rather than requiring the user to select the center, and this method does not require any predefined angular widths. Thus, the 2D Gaussian fitting method is not subjected to errors introduced by user-defined parameters in the way that the annular bin method is. Furthermore, the 2D Gaussian fitting method allows for determination of the error associated with the resulting scattering half-angles by comparing how the standard deviation in two spatial directions differs from an ideal, cylindrically symmetric distribution. Therefore, the 2D Gaussian fitting method allows for a more robust calculation of the scattering half-angles.

IV. RESULTS AND DISCUSSION

Figure 6 shows the calculated scattering half-angles as a function of the incident proton energy for the two graphene foils. The results from the 2D Gaussian fitting method are shown in orange. For example, at 10 keV, the averaged (in both the horizontal and vertical directions) scattering half-angles are $\sim 0.37^\circ$ and $\sim 0.45^\circ$ for foil No. 1 and No. 2, respectively. The differences between $\psi_{1/2,x}$ and $\psi_{1/2,y}$ are used to estimate the percent error at each energy; at 10 keV, the scattering half-angles in the horizontal and vertical directions are $\sim 0.38^\circ$ and $\sim 0.34^\circ$ through foil No. 1, resulting in a percent error of $\sim 10\%$. For graphene foil No. 1, at 3, 5, 7, 10, 20, and 30 keV, the percent error is 1.5, 5.2, 4.9, 10.0, 3.9, and 2.0%, respectively, resulting in an average error of 4.7%. The small errors indicate that the measured scattering distribution is nearly symmetric, which is interpreted as a low uncertainty associated with the scattering half-angles. In Fig. 6, the scattering half-angles determined from the annular bin method are shown in red. The annular bin results agree with the 2D Gaussian fitting method at higher energies (≥ 20 keV); at lower energies (≤ 7 keV),

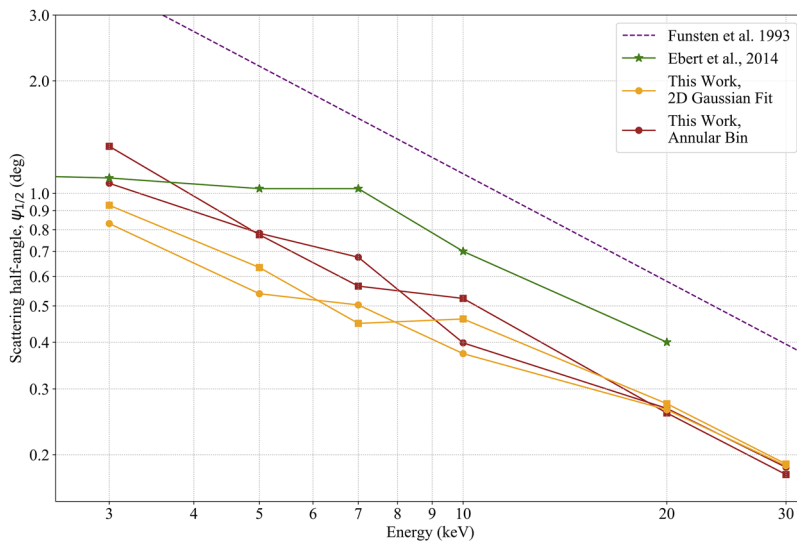


FIG. 6. Scattering half-angle $\psi_{1/2}$ of a proton beam through foils as a function of the incident proton beam energy. Analysis of the Los Alamos developed ultrathin graphene foils are shown in orange (2D Gaussian fit) and red (annular bin method). The circle and square symbols represent the calculated scattering half-angle through foil No. 1 and No. 2, respectively. The annular bin method (red) allows for direct comparison with previous measurements through graphene foils (green) (Ebert et al., 2014). The measurements through conventional carbon foils are shown in blue (Funsten et al., 1992). The small scattering half-angle $\psi_{1/2}$ of the Los Alamos developed graphene foils indicate that they are the thinnest foils reported to date.

the scattering half-angles from the annular bin method are slightly larger than those derived from the 2D Gaussian fitting method. The annular bin technique allows for a direct comparison with the previously published results through graphene foils, which used the same analysis technique. The scattering half-angles through the graphene foils tested in this study are smaller than the previously published result through graphene foils (Ebert et al., 2014), which have not been flown before, and substantially smaller than flight-proven conventional carbon foils (Funsten et al., 1993). The scattering half-angles are reduced by a factor of ~ 2.5 – 4.5 in comparison to the conventional carbon foils (Funsten et al., 1993), depending on the proton energy. The Los Alamos graphene foils reduce the

scattering half-angles by a factor of ~ 1.6 in comparison to the previous measurements through graphene foils (Ebert et al., 2014). Therefore, using the scattering half-angle as a proxy for foil thickness (Funsten et al., 1992), these foils are the thinnest foils ever produced for space plasma measurement applications.

A. Graphene foils in relation to Meyer's theory

Meyer's theory is used to describe the physics driving beam-foil interactions by developing a formula that relates the angular scattering of ions traversing a foil to the thickness of the foil. The formula is derived for small angle scattering and for $n \gg 5$, where n is the average number of scatter events encountered by each incident

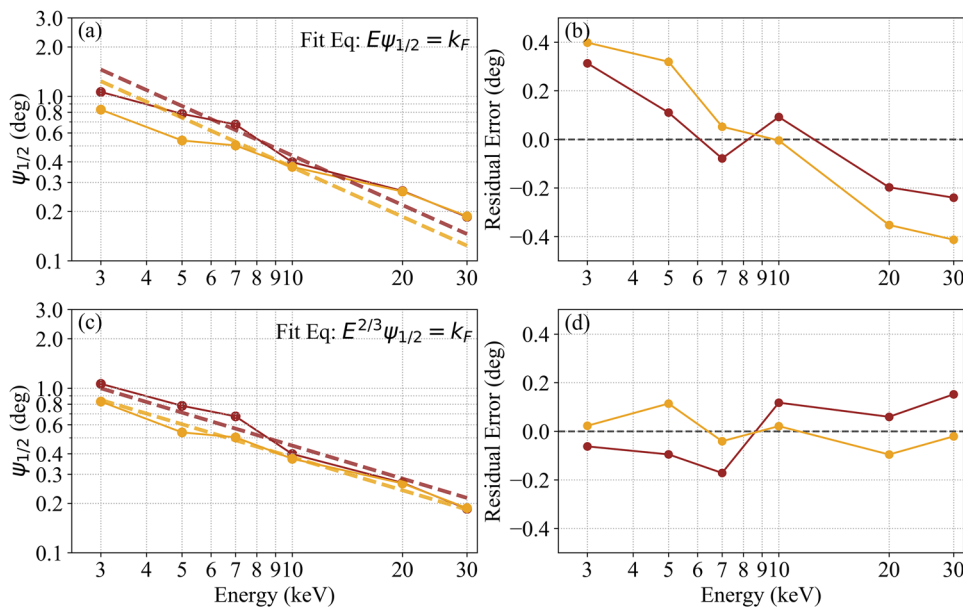


FIG. 7. Plot of the scattering half-angle as a function of incident energy shown in (a) and (c). The 2D Gaussian and annular bin methods are shown in orange and red, respectively, through graphene foil No. 1. The dashed lines are the fit to $\psi_{1/2}E = k_F$ [Eq. (7)] in panel (a) and $\psi_{1/2}E^{2/3} = k_F$ in panel (c). Panels (b) and (d) show the residuals from the fits in (a) and (c), respectively. Results indicate a $2/3$ power dependence on the energy deviates from the Meyer theory and prior measurements of thicker carbon foils.

ion (Meyer, 1971). This theory incorporates “reduced” parameterization for ion energy ϵ and scattering angle Ψ based on the screened Coulomb potential. The reduced angular half-width of the scatter distribution is

$$\Psi_{1/2} = \psi_{1/2} \frac{\epsilon(m_1 + m_2)}{2m_2}, \quad (6)$$

where $\psi_{1/2}$ is the angular half-width measured in the lab frame, and m_1 and m_2 are the atomic mass of the foil and ion beam, respectively. For a particular combination of foil composition and ion beam species, the scattering half-angle $\psi_{1/2}$ and energy incident on the foil E are inversely proportional because the reduced half-width $\Psi_{1/2}$ is uniquely dependent on the reduced foil thickness τ and the reduced beam energy is proportional to the beam energy in the lab frame ($\epsilon \propto E$). This yields

$$\psi_{1/2}E = k_F, \quad (7)$$

where k_F is a constant value for a specific combination of foil composition, foil thickness, and ion species (Funsten *et al.*, 1993).

Previous experimental measurements using carbon foils are accurately described by Eq. (7) (Funsten *et al.*, 1992; 1993; 1994; Funsten and Shappirio, 1997; and Högborg and Norden, 1970). However, the measurements through graphene foils presented in this study are *not* well-described by Eq. (7), which is consistent with previous measurements through graphene foils (Ebert *et al.*, 2014). Figure 7(a) shows the measured scattering half-angles as a function of incident energy that have been fit to Eq. (7), which predicts a stronger dependence on energy as indicated by the resulting fit residuals [Fig. 7(b)] that exhibit a clear trend. Figure 7(c) shows the same data empirically fit to a $2/3$ power dependence on energy, i.e., $k_F = \psi_{1/2}E^{2/3}$, yielding k_F values of $1.78 \text{ keV}^{2/3} \text{ deg}$ and $1.89 \text{ keV}^{2/3} \text{ deg}$ for the two graphene foils of this study. The residuals shown in Fig. 7(d) indicate a better fit to the data in comparison to Fig. 7(b). Meyer's theory is tied to the condition that there must be significantly greater than 5 collisions when an ion traverses a foil (Meyer, 1971). Because the graphene foils tested in this study are 3 atomic layers thick, we hypothesize that the condition for validity of the Meyer theory is not met and that the extremely thin graphene foils may represent a new domain of beam-foil interaction physics. Further studies can elicit insight into the breakdown of Meyer's theory for these ultrathin graphene foils.

V. CONCLUSION AND FUTURE STUDIES

The angular scattering distributions of protons through two 3 mm diameter graphene foils that are 3 atomic layers thick ($\sim 5 \text{ \AA}$ per layer, total thickness of $\sim 15 \text{ \AA}$) were measured. Two different methods were used to determine the scattering half-angle: the 2D Gaussian fit and annular bin method. Using the angular scattering measurements as a proxy for foil thickness, both analysis methods yield consistent results that the Los Alamos developed graphene foils are the thinnest foils reported to date. We find that these Los Alamos developed graphene foils are a factor of ~ 2.5 – 4.5 times thinner than the flight-proven conventional carbon foils (Funsten *et al.*, 1992) and a factor of ~ 1.6 times thinner than other new graphene foils described in the literature (Ebert *et al.*, 2014), which have not been flown before. The measured scattering half-angles are inconsistent with current theories describing ion-foil interactions. It is

postulated that the graphene foils are sufficiently thin that these traditional theories no longer hold. These thinner foils will improve the mass resolution for foil-based TOF space instrumentation by reducing angular scattering as well as energy loss and straggling.

Since graphene foils have no flight heritage to date, there are several studies that must be completed in order to prepare this new technology for usage in space:

1. It is important to characterize the unwanted affects associated with using ultrathin foils, which entails measuring the angular scattering distribution of a variety of ion species (not just protons) and characterizing the energy loss and straggling through graphene foils. This will allow us to understand the extent to which 3-layer graphene foils degrade the instrument response.
2. In some space instrumentation applications, carbon foils are used to alter the charge state of the incident ion and/or generate a secondary electron. The exit charge state distributions of ions traversing graphene foils that are ~ 3 – 7 layers thick have been previously measured; this study found that graphene and carbon foils yield similar exit charge state distributions (Allegrini *et al.*, 2014). Nevertheless, it is still important to measure the charge state distribution and secondary electron yield for the Los Alamos developed graphene foils.
3. Finally, it is important to test the durability of the graphene foils to ensure they can survive the launch and flight environments. It is important to conduct environmental testing of these foils, including vibration, shock, thermal, and thermal-vacuum characterization (e.g., McComas *et al.*, 2004).

Graphene foils are a new, exciting technology with the potential to greatly improve space plasma instrumentation performance. We have demonstrated that the Los Alamos developed graphene foils are the thinnest produced for space application. This study in conjunction with the future work outlined above is the first critical step toward preparing graphene foils for flight.

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